

Introduction to Data Science: Data pre-processing

Tutorial 4

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1. Consider a data set consisting of 50 observations in 5 different variables X_1, X_2, X_3, X_4 and X_5 . We use the notation $x_{i,j}$ to denote the i -th observation of X_j and we assume this data set is standardized. Assume that Z_1 and Z_2 can be defined as follows:

$$\begin{aligned}Z_1 &= \alpha_{1,1}X_1 + \alpha_{1,2}X_2 + \alpha_{1,3}X_3 + \alpha_{1,4}X_4 + \alpha_{1,5}X_5, \\Z_2 &= \alpha_{2,1}X_1 + \alpha_{2,2}X_2 + \alpha_{2,3}X_3 + \alpha_{2,4}X_4 + \alpha_{2,5}X_5.\end{aligned}$$

The vectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ are assumed to have norm 1.

- (a) True or false? Explain your answer. $\sum_{i=1}^{50} \left(\sum_{j=1}^5 \alpha_{1,j} x_{i,j} \right)^2 = \sum_{i=1}^{50} \left(\sum_{j=1}^5 \alpha_{2,j} x_{i,j} \right)^2$.
- (b) Explain why it might be a good idea to additionally assume that $\sum_{j=1}^5 \alpha_{1,j} \alpha_{2,j} = 0$.
- (c) True or false? $\sum_{j=1}^5 \alpha_{1,j}^2 + \sum_{j=1}^5 \alpha_{2,j}^2 = 1$.
- (d) True or false? $\text{Var}[Z_1] = 1$.

Which of these statements is always correct?

2. Consider a data set with 1000 observations. Each observation contains 10 different variables with mean zero and equal variance. We perform a principal component analysis. We determine the scores $z_1, z_2, \dots, z_{1000}$ for each observation using the first principal component as follows:

$$z_i = \sum_{j=1}^{10} \alpha_{1,j} x_{i,j}.$$

The total variance of the data set is equal to 5 and you are given $\sum_{i=1}^{1000} z_i^2 = 1253$. What is proportion of the variance that is explained by the first principal component?

- A. Smaller than 10%
- B. Between 10% and 20%.
- C. Between 20% and 30%.
- D. Between 30% and 40%.

3. The Motor-Trend-Car-Road dataset consists of data that was extracted from the 1974 Motor Trend US magazine, and comprises fuel consumption and 10 aspects of automobile design and performance for 32 automobiles (1973-74 models). There are 32 observations on 11 numeric variables:

- 1 mpg Miles/(US) gallon: more powerful and heavier cars tend to consume more fuel
- 2 cyl Number of cylinders: more powerful cars often have more cylinders
- 3 disp Displacement (cu.in.): the combined volume of the engine's cylinders
- 4 hp Gross horsepower: this is a measure of the power generated by the car
- 5 drat Rear axle ratio: higher values will decrease fuel efficiency
- 6 wt Weight (1000 lbs)
- 7 qsec 1/4 mile time: the cars speed and acceleration
- 8 vs Engine block: 0 = V-shaped or 1 = straight
- 9 am Transmission: 0 = automatic or 1 = manual
- 10 gear Number of forward gears: sports cars tend to have more gears
- 11 carb Number of carburetors: associated with more powerful engines

Because PCA works best with numerical data, the two categorical variables (vs and am) are excluded so we are left with a matrix of 9 columns and 32 rows. The data is standardized before PCA is applied and gives the following result:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
Standard deviation	2.3782	1.4429	0.71008	0.51481	0.42797	0.35184	0.32413	0.2419	0.14896
Proportion of Variance	0.6284	0.2313	?	0.02945	0.02035	0.01375	0.01167	0.0065	0.00247

- (a) What is the relation between the first row (Stand. dev.) and the eigenvalues λ_i ?
 - (b) Determine the Proportion of Variance for PC3.
 - (c) How many components should be retained according to the proportion of variance explained criterion at level 95%?
4. Let

$$X = \begin{pmatrix} X_1 \\ X_2 \end{pmatrix} \sim N \left(\begin{pmatrix} 0 \\ 0 \end{pmatrix}, R \right)$$

where

$$R = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}.$$

Assume that the distributions X_1 and X_2 are standardized and $r > 0$.

- (a) Determine the eigenvectors $\underline{\alpha}_1$ and $\underline{\alpha}_2$ of the covariance matrix R .

- (b) The results from the previous question imply that the principal components of a bivariate normal distribution are independent of the correlation coefficient r . Explain this intuitively.
- (c) What is the effect of the correlation r when performing a principal component analysis on a bivariate normal data set.

5. Consider a data set with car information.

Variable	PC 1	PC2
gas mileage (MPG)	0.3	0.02
Weight	-0.36	0.18
Length	-0.27	0.42
Engine	-0.34	0.02
Horsepower	-0.32	-0.29
Price	-0.26	-0.47

Give an interpretation of the first and second principal component.

6. Consider the random variables $\underline{X} = (X_1, X_2, \dots, X_p)$ with covariance matrix denoted by Σ . Define the variable Z_1 as a linear combination of the variables $\underline{X}$:

$$Z_1 = \alpha_{1,1}X_1 + \alpha_{1,2}X_2 + \dots + \alpha_{1,p}X_p.$$

Show that

$$\text{Var}[Z_1] = \underline{\alpha}_1 \Sigma \underline{\alpha}_1',$$

where $\underline{\alpha}_1 = (\alpha_{1,1}, \alpha_{1,2}, \dots, \alpha_{1,p})$.

7. Consider the covariance matrix Σ of a n -dimensional random vector $\underline{X}$. Define $\underline{\alpha}_1 = (\alpha_{1,1}, \alpha_{1,2}, \dots, \alpha_{1,n})$ and

$$f(\underline{\alpha}_1) = \underline{\alpha}_1 \Sigma \underline{\alpha}_1' - \lambda (\underline{\alpha}_1 \cdot \underline{\alpha}_1 - 1).$$

Show that

$$\frac{\partial f}{\partial \underline{\alpha}_1} = 2 (\Sigma \underline{\alpha}_1' - \lambda \underline{\alpha}_1').$$

Exam question

1. Consider a data set with two variables, denoted by X_1 and X_2 . You are also given the following information

$$\mathbb{E}[X_1] = 178.5, \quad \mathbb{E}[X_2] = 76.5, \quad \text{Var}[X_1] = 275.67, \quad \text{Var}[X_2] = 126.67.$$

Moreover, you are also given that

$$\text{Var}[X_1 + X_2] = 714.$$

Observation	X_1	X_2
1	160	70
2	180	81
3	200	90
4	174	65

Table 1: Data set with four observations.

- (a) (4 points) Determine the first principal component. You do not need to standardize or scale the data.
- (b) (1 point) Determine the proportion of the total variance explained by the first principal component.
- (c) (2 points) Denote by (β_1, β_2) the second principal component. Determine the variance of the second principal component.